

Justin Yu

justinyu.github.io | github.com/justinyu | linkedin.com/in/justinyu2022

Skills

Technologies: Ray, PyTorch, Docker

Areas: Distributed Training, Streaming Data Ingestion, Reinforcement Learning, Profiling and Observability

Experience

Senior Software Engineer, Anyscale – San Francisco, CA

Aug 2022 – Present

- Drove Ray Train and Ray Tune libraries to general availability as technical lead on 5-person team through comprehensive refactoring: (1) decoupled the two libraries ([RFC](#)) to fix stability and usability issues, clear tech debt, and increase feature velocity for subsequently delivered elastic training and async checkpointing/validation capabilities, (2) consolidated checkpointing backend to fix critical large model checkpointing OOMs; refactored 5-10K lines and deleted 15K+ lines of legacy code
- Designed and implemented mid-epoch resumption for distributed training data ingestion through cross-team collaboration with Ray Data team, achieving 6x faster failure recovery at 3B+ row scale; developed benchmarking framework to track performance and prevent regressions
- Drove Ray Data training ingestion roadmap by conducting user interviews, compiling requirements, delivering features including weighted dataset mixing, and fixed critical system stability issues ([blog post](#))
- Led technical engagements with 5+ enterprise customers in H2 2024: enabled \$1.2M annual cost savings through distributed data pipeline optimization (5x faster training, 7x cost reduction); drove \$3M+ in new ARR through POCs delivering 6x cost improvement and technical deep dives that converted 3 new customers
- Identified profiling gap where post-hoc code changes for profiling were too slow; implemented zero-code GPU profiling by integrating dynolog into cluster setup and training dashboard for on-demand trace collection; adopted by multiple customers for diagnosing training bottlenecks
- Wrote extensive documentation and examples enabling customers to successfully distribute their training workloads with Ray Train and Ray Data ([docs](#)); presented [Distributed Solutions for Removing Training Bottlenecks](#) at Ray Summit 2025
- Onboarded 6 engineers through structured projects and mentored 2 junior engineers; reviewed 100+ PRs per half; led paper reading groups and training sessions with field engineers
- Ray committer and SkyRL contributor

Software Engineer Intern, Two Sigma – New York, NY

May – Aug 2021

- Built multi-agent reinforcement learning system using Ray and RLlib to model strategic trading based on market microstructure theory; compared learned RL policies against analytical equilibrium solutions for single-step case
- Compared RNN, linear, and constant baseline trading policies for multi-step trading through ELO ranking; demonstrated RNN policies outperformed by leveraging price history

Undergraduate Researcher, [Robotic AI & Learning Lab](#) – Berkeley, CA

Feb 2019 – May 2022

- Built vision-based reinforcement learning system for learning robotic manipulation tasks without manual resets; implemented algorithms, conducted simulated and hardware experiments on three-fingered robot hand, and performed hyperparameter tuning and ablation studies; co-authored [ICLR 2020 spotlight paper](#)
- Extended the system to the multi-task setting where learned policies compose into self-resetting behavior; designed task graph framework allowing different tasks to reset each other, enabling scalable autonomous training; co-authored [ICRA 2021 paper](#)

Education

University of California, Berkeley – B.S. in Electrical Engineering and Computer Science

May 2022

GPA: 3.97/4.0

Teaching Assistant for EE16B (Intro EE and Linear Algebra) and CS189 (Machine Learning)